Ng'ombe: a recorded herd

A records pilot for the Southern Province cattle sector: the mandate and the funding requested

DRAFT, version 0.84, 7 November 2026, not yet ready for submission · to be submitted by Zamai Solutions, Batoka, Choma district, with Matthew Sakala and Dr Enock Siankwilimba



ngombe.zamai.pro
On paper: scan to open the demo on a phone. On a phone: tap the address.
Chrome or Edge, then "Install on this phone".
Questions: Christoph Lindinger, +260 96 2653100 · chris@zamai.pro

1. The decision brief

Ng'ombe, and the Tonga. Among the Tonga of the Southern Province cattle have never been only livestock. They are the household's bank and its standing: bridewealth is paid in them, a funeral is marked with them, a dispute is settled with them, and a man or a woman is measured by the kraal, the ciba, that stands by the house. The plough oxen that made the Plateau Tonga the country's leading smallholder maize growers a century ago were cattle, and the herd that a family loans out to kin across the villages is a web of obligations that an outsider cannot see. The language carries this: Tonga has its own words for the colours, the markings and the horns of an animal, for its age and its standing in the herd, for the acts of herding, dipping, loaning and selling; these are words that children learn at the kraal and that have no equal in English. The province holds more cattle than any other in Zambia, about 1.75 million head in the 2024 survey. A records service for these herds is therefore not only a business tool. Built in Tonga, by Tonga speakers, it keeps the words of the kraal in the technology that a young herder carries in his pocket, where otherwise the English of the phone would replace them. That is why the line reads Tonga, why every fixed line a herder sees comes from a speaker, and why the people translating and reviewing this work are named on its cover. The introduction you are reading is to be read first in Tonga, in Matthew Sakala's rendering with Dr Khama Hang'ombe's review; the English stands here until it is in.

The proposal. A one-season test of a cattle records service under the veterinary service, wherever it chooses to try it; what we have and what we suggest is on page 3. One record per animal, written by the people who know the animal: the herder, the owner, the veterinary assistant at the camp, the abattoir. A sick report reaches the veterinary assistant and the district veterinary officer with the photo and the animal's history, in Tonga or English; the carcass result comes back to the farmer. Nothing more in the first season: no cameras, no breeding programme, no models.

The mandate requested. That the Provincial Veterinary Officer and the Director of Veterinary Services appraise this proposal within sixty days and, if approved, give the pilot its mandate in writing, in five lines: the area and the period, named by the ministry; an authorised departmental lead with decision rights and a reporting line, to whom we report monthly; the veterinary assistant at the chosen camp and the district veterinary officer in the reporting loop, with the response protocol theirs to approve; access, meaning the officers' time, the dip day schedules, permission to photograph and

record animals at the tank, and the camp's existing records where they help; and an agreed arrangement with the national identification and traceability unit on identifiers and data exchange before the first record is written. A letter from the Delivery Unit opens the door; the mandate that reaches a camp officer is the Director's.

What we commit to. The development work: the records system adapted to cattle, the herder's and the camp's screens, the WhatsApp line reading Tonga, the training material written with the camp, hosting on our own machine, and a monthly report with the numbers, good or bad. That is ours and it is contributed. Where it is tried, with whom, how large and for how long is not ours to decide or to commit: page 3 says what we have and what we suggest, and the partners decide. No farmer pays, in the pilot or after. This work is not for profit for as long as it is used to promote smallholder cattle farming in Southern Province; any commercial use, by a processor for instance, is discussed separately.

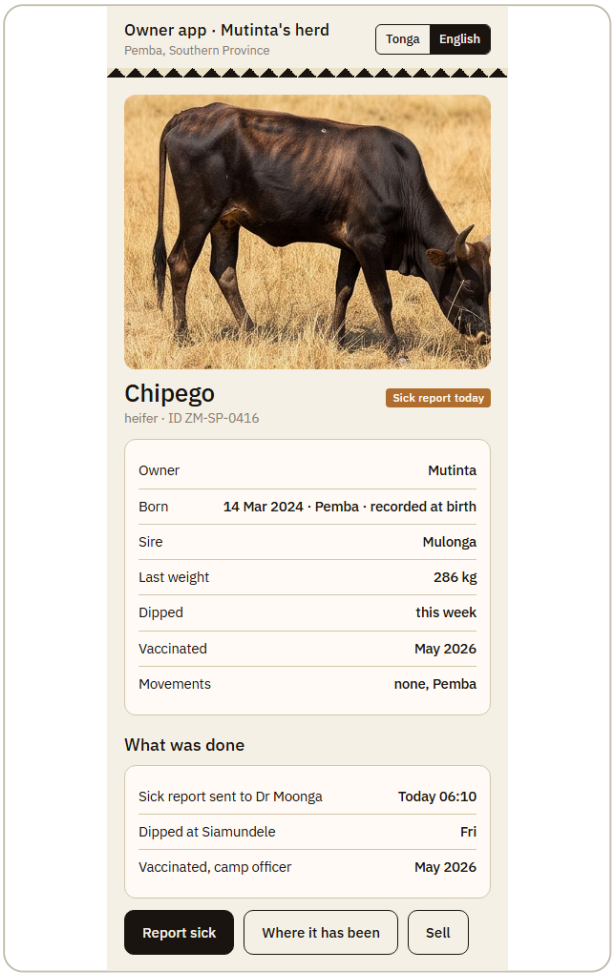
What it costs, and what we seek. The software is not the cost; we carry it. The cost is people in the field, the production of data and the machine the records live on. Two phases over two years, about 215,000 US dollars indicative, from the partners or from a funder the partners help us approach, with our own contribution in kind; the tables are in section 3. Once it works, running it costs about 5,200 US dollars a year at pilot scale, people in the field aside: a record costs nothing to keep.

What works today is in section 6, under what is built and running; what remains to be agreed is in this brief.

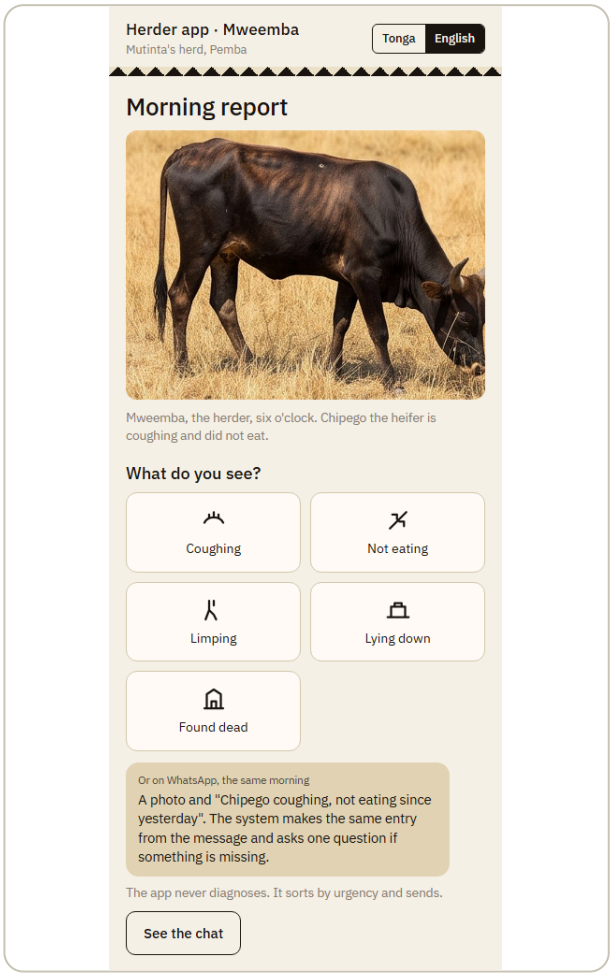
The test, and the exit. We would suggest that a test continues past its first season only if sick reports are answered within an agreed time, if records are complete at the point of sale, and if herders are still using it in month six without being chased. If the season shows that information was not the missing input, we say so, and the records stay with the camp in a form it can use.

2. What the pilot does, screen by screen

The demo follows one herd near Pemba for a week. The animals, prices and provincial figures in it are demonstration data. Four screens carry the pilot's whole workflow.



One animal, its record: owner, date of birth (witnessed where the animal was registered at birth, otherwise the owner's estimate or an age range from the teeth), sire where known, weight, dipping, vaccination, movements. The official identifier is the animal's name in the system.

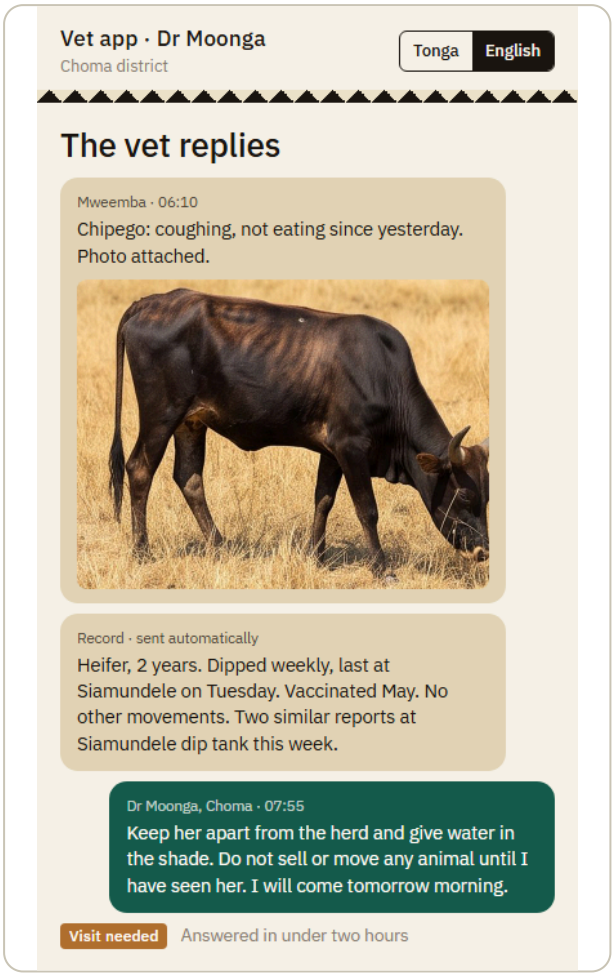


The herder's report: tap the signs, add the photo. Cough and not eating together are flagged urgent under triage rules the veterinary service approves before the pilot. "Found dead" carries the approved precaution: do not open or move the carcass, keep people and dogs away, report at once. The app never diagnoses.

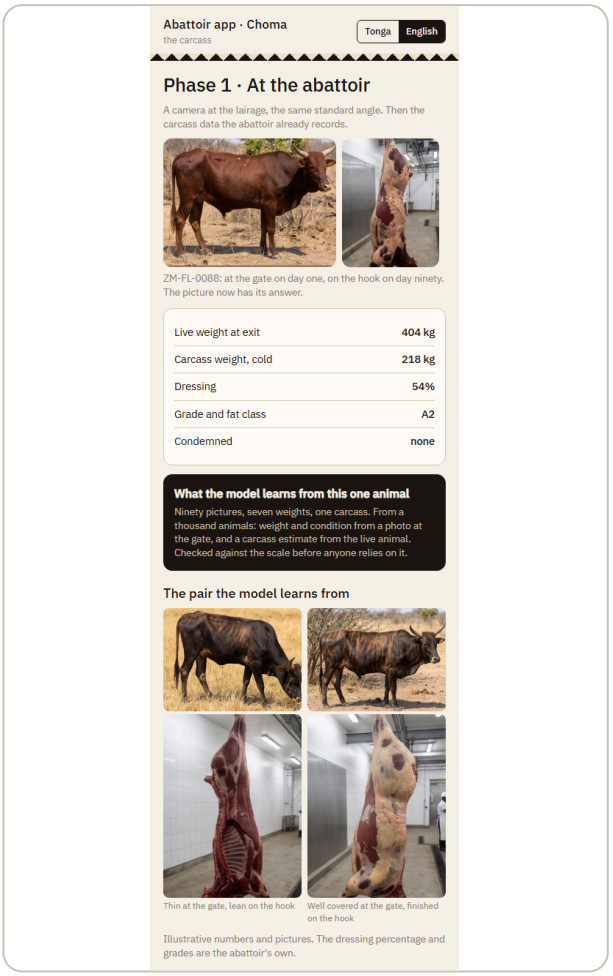
Between these screens sit the entries the pilot collects: dip day as a list of animals present, a movement entry at the dip tank and the sale pen, the vaccination as product, batch, date and the person who gave it, a death with its cause as the vet recorded it. Three reports with the same signs from three different animals in herds using one dip tank within seven days raise a cluster alert to the district veterinary officer. That alert sits on top of the Animal Health Act's duty to report suspected disease at once, never instead of it; the first report goes to the vet immediately.

What the record gives, in the first season. A report that reaches someone responsible for responding, with an acknowledgement and a recorded outcome. A record of dipping, vaccination and

deaths by animal instead of by count. The carcass result back to the farmer. A reference price from pooled sales, published as aggregates, once there are enough recorded sales to pool.



The veterinary reply, with the photo and the animal's history in front of the vet. One fictional response to one assessed case; the response protocol, hours and cover are the veterinary service's to set.



The abattoir closes the loop: carcass weight and grade linked to the animal's identity, returned to the farmer. The first pair of live weight and carcass result is what any later weight-from-picture model would learn from.

What comes later, if the records exist. Movement histories that the veterinary service can use in an export assessment under the applicable pathway; recorded herds that a cooperative or a lender can act on; breeding records with the farmers' own objectives.

3. What we have, and what we suggest

What we have is the technology, and the people who built it. Where it is tried, with whom, how large and for how long is the veterinary service's and the partners' decision, not ours. What follows is a suggestion, so that the appraisal has something to correct.

What we offer, wherever it is wanted. The records service as it stands: profile, terms and a phone tied through the line; animals with pictures; sick report, dip day and the other entries by app or by WhatsApp, in English or Tonga, with no language setting to choose; the record that travels to an abattoir; the owner's and herder's screens; the camp's screens to follow. It can be tried at a veterinary

camp, a dip tank area, a cooperative, a feedlot, which the partners have and where the gate-to-hook pair is easiest to record, or an abattoir, in any combination the partners choose.

What we suggest. One place and one season, the records service only, with a comparable area alongside where nothing changes. Measured with denominators: sick reports received, acknowledged within an agreed time, and closed with a recorded outcome; the share of enrolled animals with dipping, vaccination and movement entries complete at the point of sale; deaths and calvings recorded per hundred enrolled animals, read first as better counting; the share of enrolled herders still making entries in month six without prompting; the cash cost per enrolled animal. Suggested gates: if by month three fewer than half the enrolled herders have made an entry, or reports go unacknowledged, stop and say why. One place and one comparison area can show whether the service is used and whether reports are answered; they cannot show that it changed mortality, and we would not claim it.

Contributed by Zamai, and committed: the development work, its hosting on our own machine, the monthly report, the adaptation of the responder to cattle, and our time. Whatever funding arrives pays the contributors' ledger first: the Tonga and field work done unpaid to get here. The Tonga is contributed from the partners' side: Matthew Sakala is already on it. The staff time of the veterinary service and the cash costs are the partners' to decide and carry.

What it costs, and what we seek. The software is not the cost: it exists, and we carry its development and hosting ourselves. The cost is the people in the field, the production of data, and the machine the records live on, none of which can be copied and all of which have to be paid for. Two phases, indicative, in US dollars, each line costed at the upper end of what we expect and 15 percent contingency on top: a first season at one or two sites, 64,400; then the three models and readiness for a province over the following fifteen months, 151,800. Over two years, 216,200, about 215,000. We seek this from the partners, or from a funder the partners help us approach, and we bring our own contribution in kind: the development, the hosting, a year of records practice behind it, and the time of the people who built it. The people who have worked unpaid to get this far, the translator and the field contributors, are paid first from whatever arrives.

What the whole of it needs, two phases, indicative, to be priced

Phase 1, one season, one or two sites, about nine months	US$
A field and data lead, part to full time	8,000
Two field recorders for dip days and scoring support	7,500
The Tonga desk: the app, the line's replies, checking	3,000
2,000 to 3,000 scored pictures: 25 to 40 dip days with a scorer and a recorder, transport	6,000
Phones, chargers, power banks; a server, its power backup and connectivity for the year	6,500
Model training runs and the line's model calls	2,000
Two motorbikes for the recorders, with helmets, registration and insurance; fuel and maintenance for the season	7,000
Field days at K500 per person for food, airtime and an allowance; a bed when the tank is far; a vehicle at daily rates when a bike will not do	5,000
Training days and meetings; data-protection registration and agreements; measurement and the report	11,000
Subtotal	56,000
Contingency, 15 percent	8,400
Phase 1, indicative total	64,400

Phase 2, the three models and readiness for a province, months ten to twenty-four	US$
The same team for fifteen more months, plus one junior Zambian developer	50,000
Pictures to 5,000 and more, three areas, two seasons	12,000
Tonga speech: thirty to fifty hours transcribed, one voice recorded	10,000
Training runs for the text, visual and audio models	6,000
Five to ten more sites: phones, one or two camera sites at tanks	20,000
Two more motorbikes for new recorders, with gear, fuel and maintenance for fifteen months	9,000
Logistics across the province: a vehicle at daily rates and fuel, per diems and accommodation for three areas over two seasons, airtime and data	15,000
Training and meetings across the province	10,000
Subtotal	132,000
Contingency, 15 percent	19,800
Phase 2, indicative total	151,800

What it costs to run, once it works. The build is the cost; the running of it is not. At pilot scale the service needs the machine's power backup and satellite connectivity, the machine itself written off over three years, the line's model calls at under one US cent a message, one refresh of the models a year, and hosting and domains: about 4,500 US dollars a year, 5,200 with 15 percent contingency. People in the

field are a cost only while data is being produced; the Tonga desk only when new words are needed. A record, once the system exists, costs nothing to keep: at ten thousand animals the running cost is about fifty US cents an animal a year, at a hundred thousand a few cents.

Each line is the upper end of what we expect, so that the total holds. Rates assumed: field staff at K6,000 to 20,000 a month by role, a motorbike at about K30,000 with fuel and maintenance about K3,000 a month, a field day at about K500 per person for food, airtime and a small allowance, vehicles at daily rates when needed, a junior developer at K10,000 to 15,000, the desk's existing per-line rates for Tonga, rented GPU hours for training, under one US cent per message for the line's model. Nothing in these tables is a commitment; the partners price what they carry, and a funder's own template will reshape the lines.

4. The economics, an illustration

An illustrative herd under assumed annual rates: twelve cows exposed to breeding in a herd of thirty, at government's published programme baseline and target (Presidential Delivery Unit, livestock value chains page). The IAS household average in the province is about ten animals; thirty is chosen to make the arithmetic visible. K8,000 a head is a working value, the same for every animal. It is about right for the herd as a whole: the province's household herd is 39 percent cows, 19 percent oxen, 15 percent heifers, 11 percent calves, 7 percent bulls, 9 percent weaners and steers (ZamStats, 31 December 2024); at per-class values from an abattoir price of about K55 a kilogram of carcass (an ox or local bull about K11,000, a breeding cow K9,500, a heifer K7,500, a weaner K4,500, a calf K2,500) the herd-weighted average is about K8,500, in a range of K5,500 to K13,000 on that unpublished price. Nothing here is a forecast of this service's effect.

	Baseline	Target	Difference
Calving rate, cows exposed	29%	40%	+11 points
Calves born	3.5	4.8	+1.3
Mortality, whole herd, a year	12.5%	5%	-7.5 points
Animals lost	3.8	1.5	-2.3
Births minus deaths, before sales	-0.3	+3.3	+3.6 head
Head-equivalent value at K8,000, not cash, not profit			about K29,000

At the baseline the herd stands still: the calves born barely replace the animals that die. At the target the natural increase is about a tenth of the herd a year, before sales and before the time it takes a calf to become a sale animal. What moves a herd from one column to the other is grazing and water through the dry season, dipping, vaccination, a vet who answers, and a bull that works. Records make those actions visible and timely; they do not replace them.

For the province, the scale of the opportunity: 1,747,550 cattle at 31 December 2024 (ZamStats), of which 78.5 percent in small and medium households. One percentage point of mortality is about 17,500 animals a year. If mortality fell from the programme baseline of 12.5 percent to its target of 5 percent

across that population, about 131,000 fewer animals would die a year, about K1 billion in head-equivalent value, derived. That is the size of the target, not the expected effect of a records pilot.

The export target, worked backwards. Zambia's own beef exports were 971 tonnes in 2025, nearly all frozen to the Democratic Republic of Congo at US$3.79 a kilogram, up from 371 tonnes at US$1.58 in 2023 (UN Comtrade, reporter Zambia); the Gulf and the EU pay more, as section 5 shows. At US$4 to 8 a kilogram of carcass-weight equivalent, a billion dollars is 125,000 to 250,000 tonnes, 0.6 to 1.1 million head a year at 220 kilograms a carcass, 5 to 10 percent export offtake on a herd of eleven million. The target is a thousand times today's exports, and every one of those animals needs a history. The export offtake comes out of the same natural increase that domestic slaughter and replacement heifers come out of, so the national herd target needs its own herd-flow model; this note does not supply one. What the export sum does say is that a million export animals a year need a million histories, and the first entries in each are made at the kraal and the dip tank.

5. The market, and what records add per animal

Sourced in our research note of 7 November 2026 (93 sources); the figures below carry their dates.

The markets. The Democratic Republic of Congo is the market Zambia has: 971 tonnes of frozen beef in 2025 at US$3.79 a kilogram, up from 371 tonnes at US$1.58 in 2023, nearly all of Zambia's beef exports (UN Comtrade, reporter Zambia). It asks for a veterinary certificate and a licensed abattoir, not a disease-free zone; South Africa was shut out of it in February 2026. The Gulf, Saudi Arabia first, is the market to work toward: 141,000 tonnes of frozen imports at US$4.44 a kilogram and 55,000 tonnes chilled at US$6.20 in 2024, bought from India, Pakistan and Ethiopia as well as Brazil and Australia; it needs an approved plant and halal certification per consignment, not a zone. The European Union pays the most, about €6.35 a kilogram of carcass, and is closed to the region: Botswana lost its EU access in April 2026 and Namibia in October, both to foot-and-mouth disease. Zambia cannot get there without a recognised zone, the 2024 identification Act in force and listed abattoirs. China is capped by a 55 percent safeguard tariff above country quotas through 2028.

The grade that matters is South Africa's classification, read by every neighbour: age from the permanent incisors (A, AB, B, C), fat from 0 to 6, conformation from 1 to 5. It needs no marbling, and class A sells about 28 percent above class C per kilogram. Zambia has no published classification. Half the carcass value sits in a third of its weight, the rib and the loin; chilled primals fetch 1.4 to 3.4 times frozen in the Gulf and China.

What records add per animal, in order, from the neighbours' and Australia's grids

What the record proves	What it is worth	Where measured
Certified origin and movement, which opens a market	10 to 35 percent	Botswana's EU-eligible price against the ordinary one; Zambia's own unit value more than doubled 2023 to 2025
Age class from the teeth, confirmed by the record	5 to 10 percent on young animals	South African and Botswana grids
Eating-quality compliance	about A$0.36 a kilogram, A$110 a head	Australia, 2021 to 2023
Defects avoided: dark cutting, bruising, condemnations	2 to 5 percent on average, up to A$90 on an affected carcass	Australia, Namibia

On a 220 kilogram carcass at the 2025 DRC price, about US$836, that is roughly US$80 to 290 for access, US$40 to 80 for age, US$50 for quality and US$15 to 40 for defects avoided, with another US$190 to 380 if about 50 kilograms of rib and loin go chilled at two to three times the frozen price. All of it is derived from foreign grids; none of it has been measured on a Zambian animal.

The caveats. The identification Act is not yet in force. Foot-and-mouth has been in Western Province since April 2026, and the neighbours show that one outbreak closes a market for months. The premiums above are paid by buyers with price grids; the DRC has no grid yet, so the first gain for Zambia is access, not grade. The one measurement that replaces the borrowed numbers is one season of carcass pairs at one abattoir: the live record against cold dressed mass, class and condemnations. That is already on the spec sheet, and it is what the records service exists to produce.

6. Readiness, data, law

Built, running, to be built. Running since 7 November 2026: the records service on our own machine, with row-level access, hash-stamped entries, terms accepted before the first record, the phone tied through the WhatsApp line, animals with pictures, the owner's export; the line filing codes, photos, reports on a named animal, dip day, deaths, sales, calves and sick reports within seconds, English, Tonga or mixed, the Tonga read through our translator and kept with the original; the Tonga translator and desk; the demo. To be built: the camp's screens and the veterinary reply; movement entries; the abattoir entry; the line's replies in Tonga (from a speaker, about twenty hours of teacher input, Matthew Sakala is on it); a vet in the loop on the line. To be agreed: the triage rules, the response protocol, the minimum record, the identification arrangement. Research, not in the pilot: weight from a picture, the muzzle as identity, breeding evaluation.

Feasible, and beyond what we can fund internally. Three models sit behind this service, and each is within reach technically: a text model that runs on a local machine and reads the herder's messages, so that no message leaves the country; a visual model that reads an animal's body condition from a picture, and later its identity; and an audio pair that listens to spoken Tonga and answers with a voice note, for the herder who does not read. Training them is not the constraint; the data is. A visual model needs a few thousand pictures of different animals from different angles, seasons and places,

each scored by someone who knows cattle, which is a season of dip days with a recorder and an officer at the tank. The audio pair needs tens of hours of Tonga speech transcribed by speakers, and a few hours of clean recorded Tonga for the voice. The line already stores every picture and every voice note against the animal and the date, so the collection can start with the pilot; the scoring, the transcription and the recording are paid work for Tonga speakers and cattle people, and that is what we cannot carry on our own. We state it here so that the appraisal sees the whole path, not as a request.

It understands Tonga, by design. The herder writes Tonga, the line reads it, files what was said and answers in plain words; the Tonga it reads is checked by the teachers' desk, which corrects what it gets wrong. The same approach won the agriculture track of the World Bank's Small AI for Development hackathon in October 2026, an SMS cooperative for coffee farmers in Timor-Leste reading Tetum; we had built ours months earlier for Yumi Milling's WhatsApp line, where it answers staff and customers daily. Every health line a farmer reads comes from a Tonga speaker, not from the model.

The law the pilot works under. The Animal Health Act No. 27 of 2010: suspected disease is reported at once under section 17; movement permits under section 16 and export permits under section 29 are the veterinary service's, and no entry in this system authorises a movement or an export. The Animal Identification and Traceability Act No. 2 of 2024 establishes the identification and traceability unit within the veterinary service, official marks, premises and records; its commencement by statutory instrument is confirmed with the ministry before the pilot describes any requirement as operative. The Animal Health (Amendment) Act No. 7 of 2025 directs the Director to use that system for disease control and traceability, which is why the pilot's records must sit in an agreed relation to it from the start: identifiers, which events the pilot may submit, who validates them, which system is authoritative, and how the two are reconciled, settled with the national unit, not assumed. By November 2024 the identification programme had registered 88,211 farmers and 529,266 animals against a five-million target due that December, and revised the target to three million by 2027; a later report distinguishes 580,332 cattle tracked through branding from 6,102 individually ear-tagged in the Kazungula pilot. The current position is obtained from the unit before launch.

Whose data. The records contain personal data wherever they identify a person, directly or by number, kraal or location, and the Data Protection Act No. 3 of 2021 applies. Before the first record: the controller and processor named and registered; the lawful purposes stated, separating statutory disease reporting from optional uses such as model training; access by role and purpose, so that a routine provincial summary and an authorised disease investigation have different access; correction, retention and erasure handled lawfully alongside the append-only ledger; the WhatsApp channel and any rented training computing mapped against sections 70 and 71 on processing and transfer; and an impact assessment if section 46 requires one. The owner sees and corrects her animals' records and decides on uses beyond the statutory ones. No farmer pays, in the pilot or after: that is a principle, not a pilot concession. Ownership of any future models is settled in the spec sheet, not here.

The veterinary chain, as it is. Veterinary assistants at camp level, district veterinary officers, the provincial veterinary officer, the Director. Reports go to the camp's veterinary assistant, cluster alerts to the district officer; the protocol for hours, urgency, escalation and unacknowledged reports is theirs to approve, with transport, sampling and laboratory access part of it.

7. Who we are, and the references

Zamai Solutions is the applied-AI work of Yumi Milling Ltd, a maize mill in Batoka, Choma district, in Southern Province since 2019; Christoph Lindinger, its director, leads it. We build records systems, made the Tonga translator that teachers in the province test, and run the WhatsApp line that files records daily. Matthew Sakala, Choma, +260 97 7668795, is a partner in this proposal: he contributes the Tonga and brings it to the people who decide. Dr Enock Siankwilimba, +260 96 9832511, is a co-submitter: a PhD in business and management from the University of Zambia on smallholder cattle business models, twenty years in agricultural extension, now an independent adviser; his place in this proposal is being agreed with him before submission. The cattle work is not for profit for as long as it is used to promote smallholder cattle farming in Southern Province: no subscriptions, no charge to any farmer. Any commercial use, by a processor for instance, is discussed separately. Contact: Christoph Lindinger, Director, Yumi Milling Ltd, Batoka · +260 96 2653100 · chris@zamai.pro · WhatsApp on the Zamai line, +260 776 621 255 · the demo: ngombe.zamai.pro.

What is shown, and what stays with us. The app, the demo, the database design and these documents are open to the appraisal and public. The methods by which the line reads, decides and files were built over a year on a working business system in Batoka and are ours; the partners get them through the service. We say this plainly because an idea on paper is easy to copy and the work behind it is not.



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Settled before launch: the triage rules and response protocol; the identification arrangement; the data controller, purposes and access; the dip tank at the chosen site; who carries the cash costs.
Investigated by the test: use without chasing; reports answered in time; records complete at sale; cost per animal.

References. Our note *Beef markets and grading, 7 November 2026* (93 sources, UN Comtrade and the importing markets' own statistics among them). ZamStats, 2024 Integrated Agricultural Survey, tables 5.1 to 5.4. Presidential Delivery Unit, livestock value chains and beef export pages. President's address to the National Assembly, 9 October 2026. Parliament, Committee on Government Assurances, 2025 reports, pp. 25-26 and 84-85. Agriculture Committee, dip tank inventory 2023, p. 18. Shanzuwa et al. (2026), Mazabuka. The four Acts cited in section 6. Derived figures are ours (working value K8,000 a head).